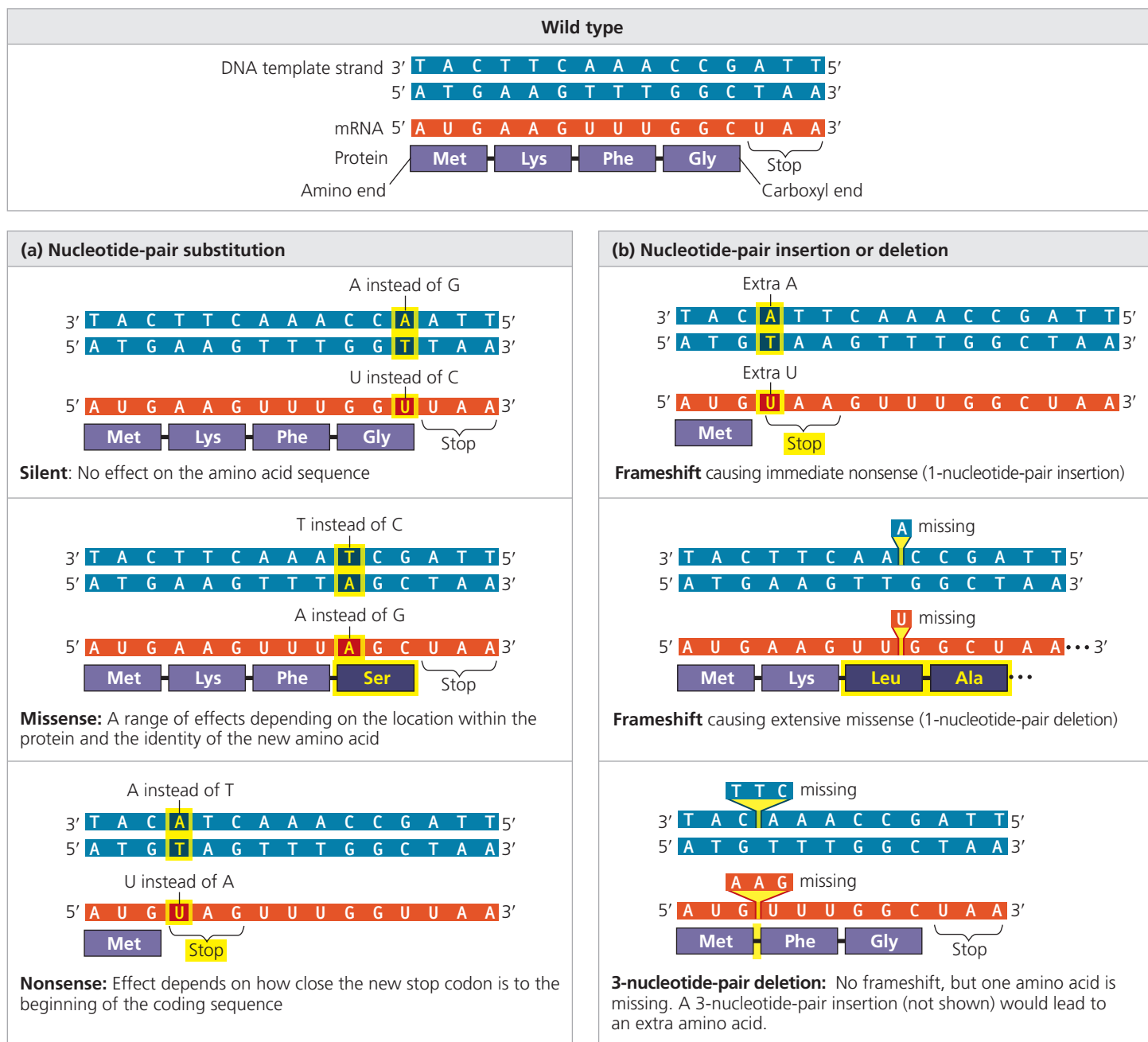


▼ **Figure 17.27 Types of small-scale mutations that affect mRNA sequence.** All but one of the types shown here also affect the amino acid sequence of the encoded polypeptide.



➔ **Mastering Biology Figure Walkthrough • Animation: Mutation Types**

hemoglobin shown in Figure 17.26 or in the active site of an enzyme as shown in Figure 8.19—can significantly alter protein activity. Occasionally, such a mutation leads to an improved protein or one with novel capabilities, but much more often such mutations are neutral or detrimental, leading to a useless or less active protein that impairs cellular function. A second example of a missense mutation is a mutation in the tyrosinase gene that causes the albino phenotype in the Asinara donkey. Recently, Italian researchers sequenced the tyrosinase gene in donkeys from this wild population and showed that a recessive mutation is responsible for the lack of pigment. A C is changed to a G in this gene,

resulting in a histidine being added to the protein instead of an aspartic acid. The changed amino acid is in a site in tyrosinase that normally binds a copper atom, and it makes the enzyme incapable of binding copper, rendering it nonfunctional. As in the case of sickle-cell disease, just one change in a nucleotide base results in a dramatic phenotypic change.

Substitution mutations are usually missense mutations; that is, the altered codon still codes for an amino acid and thus makes sense, although not necessarily the *right* sense. But a point mutation can also change a codon for an amino acid into a stop codon. This is called a **nonsense mutation**, and it causes translation to be terminated prematurely; the